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Training & Design Support Platform

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Abstract

Our work was carried out within the company Infineon Technologies AG as a part of the graduation project to obtain the engineering diploma in telecommunications at the Higher School of communication of Tunis.

This report reveals the work done at Infineon Technologies AG and which consists of creating content to Infineon customers, partners, and engineers. A tutorial, that introduces them to the radar world easily and simply. As radar becomes miniaturized and more affordable, our educational content helps the radar marketing team to advertise their product to get into new consumers and industrial applications.

The project consists in rebuilding MakeRadar webpage at the Infineon website, add new interactive application for presence detection and radar equation.

Keywords: Sense2GoL, Radar market, 24GHz Radar, presence detection, Radar equation, React, Java Script, Arduino.

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Abbreviations

CAGR: Compound annual growth rate

PIR: Passive infrared sensors

IT: Information Technology

RMS: Root-Mean-Square

MMIC: Monolithic Microwave Integrated Circuit

CMS: Content Management System

MVC: Model-View-Control

JS: JavaScript

CNN: Convolutional neural network

1D-CNN: one dimension convolutional neural network

Tx: Transmitter

Rx: Receiver

HTML: Hypertext Markup Language

General introduction

The global sensor market was valued at $138,965.0 Million in 2017, and is projected to reach $287,002.0 Million by 2025, growing at a CAGR of 9.5% from 2018 to 2025. Sensors are devices that detect events or changes in the environment and then provide the corresponding output. They sense physical input such as light, heat, motion, moisture, pressure, or any other entity, and respond by producing an output on a display or transmit the information in electronic form for further processing. These define major applications in flood & water level monitoring systems, environmental monitoring, traffic monitoring & controlling, energy saving in artificial lighting, remote system monitoring & equipment fault diagnostics, and precision agriculture & animal tracking [1].

The sensors have penetrated in almost every industry vertical but have highest usage in consumer electronic products, followed by automotive then IT & telecom. Smartphones incorporate sensors such as accelerometers, gyroscopes, ambient light sensors, proximity sensor, and temperature sensors, to keep a track of parameters and provide a centralized system for automatic control [1].

The key companies profiled in the sensor market report are STMicroelectronics, NXP semiconductors N.V., Infineon Technologies AG, Qualcomm Technologies, Inc., Atmel Corporation, Texas instruments Inc., Robert Bosch GmbH, Johnson Controls International PLC, Sony Corporation, and Honeywell International Inc. These key players have adopted product launch and acquisition strategies to strengthen their sensor market share [1].

As radar is minimized and becomes more affordable, new use cases in the consumer segment are revealed. In this context. Infineon seeks to develop a webpage, called MakeRadar. MakeRadar is tutorial for makers, where they can test the Infineon product, develop different applications and learn the basics behind them. A marketing approach to advertise their product and in an easy way.

This report summarizes the different stages of this work.
It is organized around four chapters

The first chapter will be interested in the presentation of the host company and the project context and objectives as the first part and then introduce to the used technology and development tools.

A second chapter is devoted to the state of art and study of the Radar concept, presenting Infineon product, after that a comparative study between our board and an existing solution, and finally describe the frameworks, which Infineon sensor is based on.

In the third chapter, we go into detailing the need analysis in which we present the company needs that our solution must offer.

The fourth chapter covers and highlights our project design and different solutions that we made.

The fifth chapter is dedicated to a presentation of our developed solution that we will show to our custom

Chapter I: Project context

1. Introduction

This first chapter is devoted to the presentation of the general context of the project. The work having been carried at out Infineon technologies AG, in Munich facilities, Germany. We will begin then, with a general presentation of the host company. After that, we will describe our project to clarify its context, and its aim and the various steps identified to handle efficiently the development activities of our internship.

2. Presentation of the host organization

2.1 *Infineon Corporate summary*

Infineon Technologies AG is a German semiconductor manufacturer founded on 1 April 1999, when the semiconductor operations of the parent company Siemens AG were spun off to form a separate legal entity [2].

According to the 2018 Infineon revenues, the business segments show that 9% of the revenues came from digital security solutions, 31% from the power management & multimarket, 17% from industrial power control, and 43% came from automotive. Achieving 7,599 million Euros with a growth of 0.7% compared to 2017

Infineon employs more than 40,100 people from all over the world divided into three parts of the world: Europe, Asia/Pacific, and the Americas. with 13 R&D locations and 17 manufacturing locations.

2.2 *Technology and market segments*

As previously mentioned, Infineon has four market segments. Through this part, we will briefly present them:

2.2.1 Automotive:

The automotive world is evolving at an unprecedented pace. Today, Infineon sensors, microcontrollers and power semiconductors help car manufacturers worldwide achieve

Their ever-higher challenging safety, affordability and efficiency targets. Their industry-wide leading portfolio puts her in an excellent position on the one hand to address the systems of today and on the other hand to actively shape the transformation of the automotive industry. As the number two in the area of sensors, today they benefit greatly from the continuously increasing number of driver assistance systems.

2.2.2 INDUSTRIAL POWER CONTROL

The segment Industrial Power Control specializes in the efficient conversion of electric energy along the entire supply chain (generation, transmission and consumption) with a focus on electric drives. The applications range from the wind power turbine to high voltage direct current transmission (HVDC), energy storage systems and all the way to the refrigerator. Products from Industrial Power Control enable efficient generation, transmission and conversion of electrical energy. Today, Infineon's leading market position is based on technological leadership and on being an innovative, reliable partner for their customers.

2.2.3 POWER MANAGMENT AND MULTIMARKET

My Internship was the Power Management and Multimarket department.

The Power Management & Multimarket segment covers business with power semiconductors for power supplies, components for cellular infrastructure and mobile devices as well as with high-reliability components for application in harsh environments

Infineon is expanding their portfolio of complementary drivers and controllers. Battery-operated devices are one of the fastest-growing applications for power semiconductors. In the high-frequency and sensor space, they have established a strong technology footprint with MEMS

microphones (silicon in particular), time-of-flight sensors for 3D cameras and radar applications. they have already established very successful positions in the respective markets.

The Digital Security Solutions (DSS) segment has over thirty years' experience delivering some of the world's most challenging and large-scale digital security projects. Its success is built on the Infineon wealth of expertise in conventional smart card applications. They are transferring our core skills in payment cards and government documents to the fast-growing field of embedded security applications. As digitalization shapes more and more areas of everyday life, security is becoming a key success factor for applications across industries as diverse as computing, automotive, Industry 4.0 and smart homes. Parallel to its role as an independent business segment, DSS acts as a competence center for our other three segments, supporting their efforts to hardwire security functionality into their respective system solutions.

2.2.4 SEMI CONDICTORS

Infineon provides semiconductor products for use in powertrains (engine and transmission control), comfort electronics (e.g., steering, shock absorbers, air conditioning) as well as in safety systems (ABS, airbags, ESP). The product portfolio includes microcontrollers, power semiconductors and sensors.

2.3 *24GHz radar products portfolio*

Infineon offers a wide portfolio of mmWave radar sensors to address different customer requirements. The BGT24M/L family is the largest and highest integrated 24GHz radar transceiver family currently on the market, saving ~30 percent board space compared to discrete line ups. Infineon provides a total of four 24GHz industrial radar chips, providing a range of different transmitter and receiver channel configurations, supporting different application requirements.

The Infineon range of 24GHz industrial radar chips provide four configurations of transmit and receiver channels ensuring there is a chip to support your specific application. From basic applications such as motion detection in security systems which only require one transmit and one receive channel, through to more complex applications like 3D positioning which require two or more receive channels, our range of radar chips support all of your requirements.

The next table presents the different type of 24Ghz products:

Table 1: Infineon demo kit

Sense2GOL (BGT24LTR11 + XMC1300)	Distance2Go (BGT24MTR11 + XMC4200)	Position2Go (BGT24MTR12 + XMC4700)
› Capability to detect motion, speed and direction of movement (approaching or retreating) Precise measurement of object detection compared to PIR › Operates in harsh environments and detects through non-metallic materials › Low power mode for enhanced battery life › One of the world's smallest complete radar + MCU development kit › BGT24LTR11 – 24GHz highly integrated RF MMIC › Debug over cortex 10 pin debug connector › Integrated multiple element patch antennas	› Capability to detect distance of multiple targets › Capability to detect motion, speed and direction of movement (approaching or retreating) › Operates in harsh environments and detects through non-metallic materials › BGT24MTR11 – 24GHz highly integrated RF MMIC	Capability to detect position of multiple targets › Capability to detect distance of multiple targets › Capability to detect motion, speed and direction of movement (approaching or retreating) › Operates in harsh environments and detects through non-metallic materials › BGT24MTR12 – 24GHz highly integrated RF MMI
Main applications › Security › Lighting control › Automatic door opener › Vital sensing	**Main applications** › Drone: soft landing/obstacle avoidance › Smart toilets › Tank level sensing › Intelligent switches	**Main applications** › Drone/robots: obstacle avoidance › Security › People tracking (IoT, smart home) › Vital sensing

3. Project context and objectives

3.1 *Project background*

Today, sensing market is more oriented to automotive applications, smart lighting, military and more. However, there is a segment, which is not explored “the consumer market”. The reasons behind that are lack of knowledge about radar uses cases in this segment and the existing solutions like infrared (PIR), ultrasonic and laser have dominated the market for years, simply because they are easy use and accurate results.

The reasons behind only a few customers started using radar in the consumer field because the radar is esteemed as a complicated technology that requires an RF expertise, a lot of R&D and they are more expensive.

The advantages of radar compared to current solutions are the application flexibility, higher accuracy and sensitivity, better resolution and additional features (speed detection, direction capability…). Consequently, companies are now creating new sensors with additional features and high performance to compete against existing ones.

MakeRadar tutorial is an approach to pursing Infineon radars into the consumer segment. This project has been started before in an external webpage using the following URL: www.makeradar.com. The webpage was tested for a few weeks by some of Infineon consumers and it was successful. Although, according to Infineon internal law, having an external webpage containing the Infineon brand was not allowed. The second point was related to data protection when users have to put some personal Information while ordering the product. To fix these problems and move forward, new objectives are set.

3.2 `*Project Objectives*

In this part, we will present the project objectives:

- Rebuild Make Radar tutorial on the Infineon webpage.

- Develop an interactive visualization for the radar equation
- Develop a presence detection demo for the web page and an Arduino version.

3.3 *Project methodology*

Make Radar stands for learning radar for makers. The make Radar methodology provides a structured approach to tutor about radar. It is simple and marketing-oriented methodology as we promote Infineon product through learning.

Our learning experience consists of three different levels starting with test the radar performance and goes to learning its fundamentals



Figure 1: *Make Radar Levels (customers)*

The first level, the explore part, which is composed into several applications. Each application shows one of the radar performance. The second level is called develop, this part intends to help users learn how to develop cool radar applications from level zero to become proficient, from a simple application like LED Blink to presence detection and speedometer

The make radar website does not get through all the radar details because this would make explaining it complicated.

The fourth level, develop notes, is oriented to developers of this website where they share their way of development and used tools.

4. Radar Operation Principales

4.1 *Definition*

The **RADAR** (**RA**dio **D**etection **A**nd **R**anging). detection refers to whether the target is present or not. The target can be stationary or moving, i.e., non-stationary. Ranging refers to the distance between the Radar and the target [6].

RADAR is a smart sensor, with advanced sensing capabilities that uses radio waves to determine the range, angle, or velocity of objects. It can be used to detect aircraft, ships, vehicles, pedestrians, motors etc. A radar system consists of a transmitter producing electromagnetic waves in the radio domain, a transmitting antenna, a receiving antenna and a receiver and processor to determine properties of the target(s). Radio waves (pulsed or continuous) from the transmitter reflect off the target and return to the receiver [7].

4.2 *Doppler effect*

The Doppler effect occurs when a source of waves and/or observer move relative to each other, resulting in the observer measuring a different frequency of the waves than the frequency that the source is emitting [11].

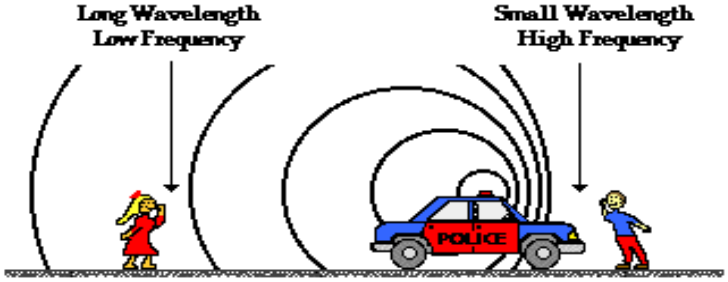


Figure 2: ***Doppler Effect for a moving sound source***

Based on the description and figure 6, we will get the following two possible cases:

- The frequency of the received signal will increase, when the target moves towards the direction of the Radar.
- The frequency of the received signal will decrease, when the target moves away from the Radar.

The distance between Radar and target is nothing but the Range of the target or simply range, R. The total distance between the Radar and target in a two-way communication path will be 2R, since Radar transmits a signal to the target and accordingly the target sends an echo signal to the Radar [11].

4.3 *IQ signals*

IQ signals in the realm of radar and electrical engineering are signals that have the same amplitude and frequency but are shifted 90° or a quarter cycle relative to each other.

Figure 7 presents the raw data of the IQ signal.

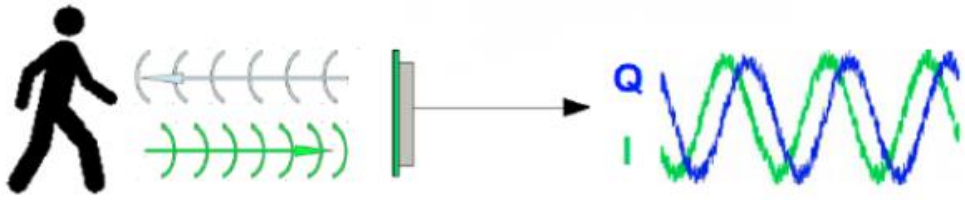


Figure 3: *IQ Signals*

Radar systems will have either real receive stages or complex ones. A real stage would only have one component of the received signal while a complex stage will generate both the I/Q components. There are several advantages of generating the IQ signals and using them for further analyses, however in the case of Doppler radar the IQ signals serve a more basic purpose [12].

Since a real mixer used in a Doppler radar receiver will give the absolute frequency shift without a direction or sign, it becomes impossible to determine the direction of travel of the target.

Using an IQ mixer, which generates both the I/Q signals, and by checking which of the signals is leading in phase the direction of travel can be determined.

5. Development Tools and Environments

5.1 *Sense2GoL Board*

Sense2GoL board is a development kit that allows the user to implement and test several sensing applications at the 24 GHz using Doppler Effect. The kit consists of the BGT24LTR11 transceiver (1Tx-1Rx) one of the smallest chip in the radar market with a MMIC and a XMC1300 32-bit ARM® Cortex®-M0 for signal processing and communication via USB [10].

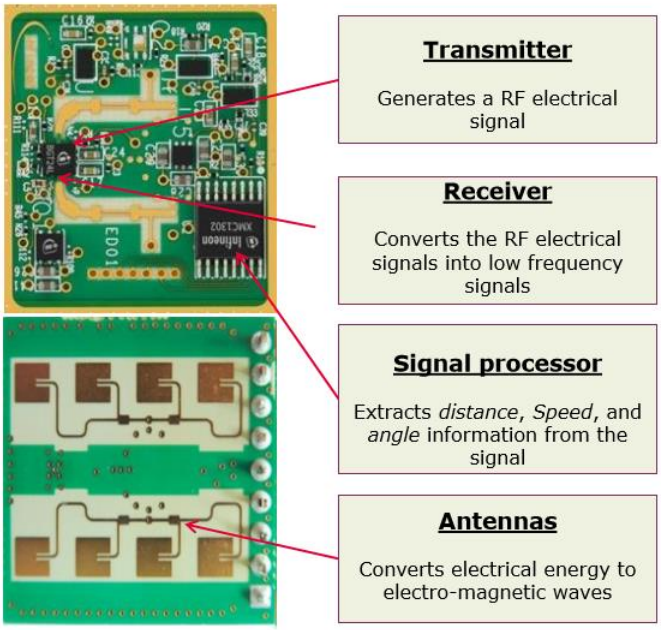


Figure 4: Sense0Gol Board

The following table gives an overview about our board:

Table 2: *Sense2GoL Overview*

operating mode	Features	Key benefits:	Target Applications:

Doppler Continuous wave (CW): - Speed measurement - Presence detection - Direction of movement (approaching or retreating)	**Maximum distance: 15 m (human target)** **Maximum speed: 20 km/h** **Radar system field of view:** **Horizontal: 29°** **Vertical: 80°**	**-High sensitivity** **-Small size** **-Can be hidden** **-Robust to environmental conditions** **-Provides speed and direction information** **Long detection rage**	**-Security** **-Outdoor lighting** **-Smart home** **-Automatic door opener** **-Intelligent switches** **-Speed measurement** **Health (vital sensing)**

5.1.1 Sense2GoL VS PIR

The next table compares Infineon 24GHz product and Infrared solution :

Table 3: *Sense2GoL vs Infrared*

Features	Sense2Gol	Infrared
Application flexibility	++	+/-
Resistance to moisture and temperature	++	+/-
Speed detection	++	+/-
Accuracy sensitivity	++	+/-
penetration of materials	++	+/-
size solution	++	++
cost	+	++

++ :Best +:Good +/-:Weak

One of the main criteria of product performance is the distance. PIR sensors are roughly 10 meters maximum, and found in places like in-home motion detectors or driveway lights.

Radar can range from within 10 meters to over 50 meters depending on the radar signature of the object. This opens the door for radar as a sensing solution for various applications where PIR traditional has limitations like hidden, sensitivity and robustness.

5.1 *Web Development Tools*

After explaining the basics of Radar and described our hardware, the next part explains the used environments to create our solution

5.2 *Open CMS*

OpenCms is an open-source content management system written in Java. OpenCms requires a JSP Servlet container such as Apache Tomcat

It is a CMS application with a browser-based work environment, asset management, user management, workflow management, internationalization support, content versioning, and many more features including proxying of requests to another endpoint.

The OpenCms user interface runs in a web browser, placing control elements directly on the web page that is edited. Content on a page can be modified by drag and drop. Text can be edited "inline" directly on the web page, or in an alternative form based editor [13].

5.2.1 React JS

a. Description

React is a JavaScript library for building user interfaces. React allows developers to create large web applications, which can change data, without reloading the page. The main purpose of React is to be fast, scalable, and simple. It works only on user interfaces. This corresponds to view in the MVC template. It can be used with a combination of other JavaScript libraries or frameworks, such as Angular JS in MVC [14].

In React, a set of immutable values are passed to the components renderer as properties in its HTML tags. Component cannot directly modify any properties but can pass a call back function with help of which we can do modifications. This complete process is known as “properties flow down; actions flow up”. Figure 9 illustrates how components are linked and flow down properties and actions flow up.

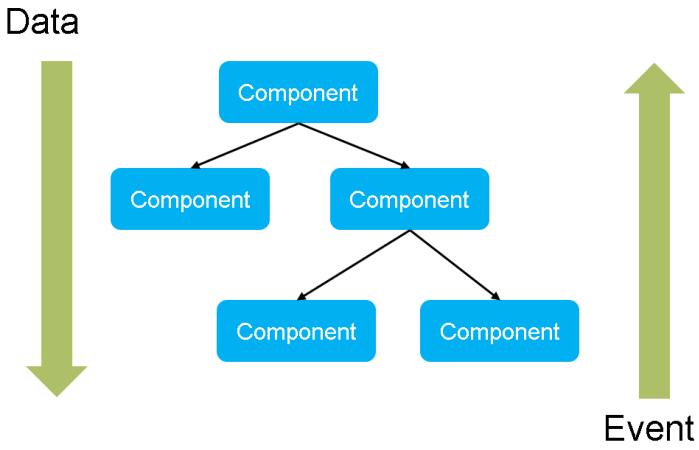


Figure 5: *properties flow down - actions flow up*

b. Structure

React JS uses a specific architecture:

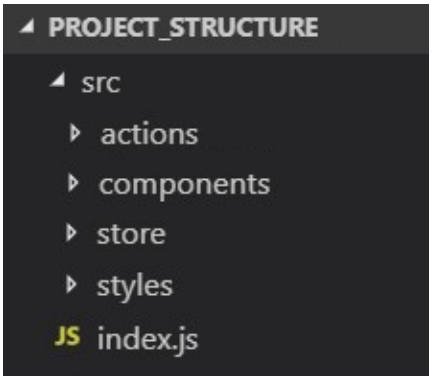


Figure 6: *React Project*

This figure shows the main elements of React JS application and we will briefly describe each one of them:

- Stores: A store holds the completely state tree of your application. The only way to change the state inside it is to dispatch an action on it.
- Actions: these are objects that should have two properties, one describing the type of action, and one describing what should be changed in the app state.
- Components: Components let you split the UI into independent, reusable pieces, and think about each piece in isolation.

5.2.2 Arduino Create Agent

Arduino Create Agent is a plugin that was designed to help Arduino enthusiasts connect their devices to the Arduino Create hub where they can quickly develop new projects, link cloud-enabled devices, access tutorials and get in touch with other members

Functionality-wise, this little tool is but a single binary that accesses data from a configuration file. It listens to HTTP and Web Socket connections. Figure 11 presents Arduino create agent



Figure 7: *Arduino Create Agent*

After launching the application, it will silently move to the System Tray and continue running in the background, without disrupting any active applications in the foreground. However, users can interact with it by right clicking the tray icon and choosing the desired feature from the combo menu [16].

6. Conclusion

In this chapter, we have defined the context of our project, the host company as well as the goal and end game of our project. Now, we move to presenting the state of the art in the following chapter.

Chapter II: Development Requirements Specification

1. Introduction

In this chapter, we will begin with a description of the radar technology, its functional characteristics. After that, we will present Infineon product and compared it to an existing solution. Thereafter, we will detail the different tools that allowed us to build our solution.

2. Radar Marketing Web site Requirements

2.1 *Website Ergonomic Requirement*

Our website must be simply designed, comprehensive and appealing for an optimal user-experience.

2.1.1 Well written texts

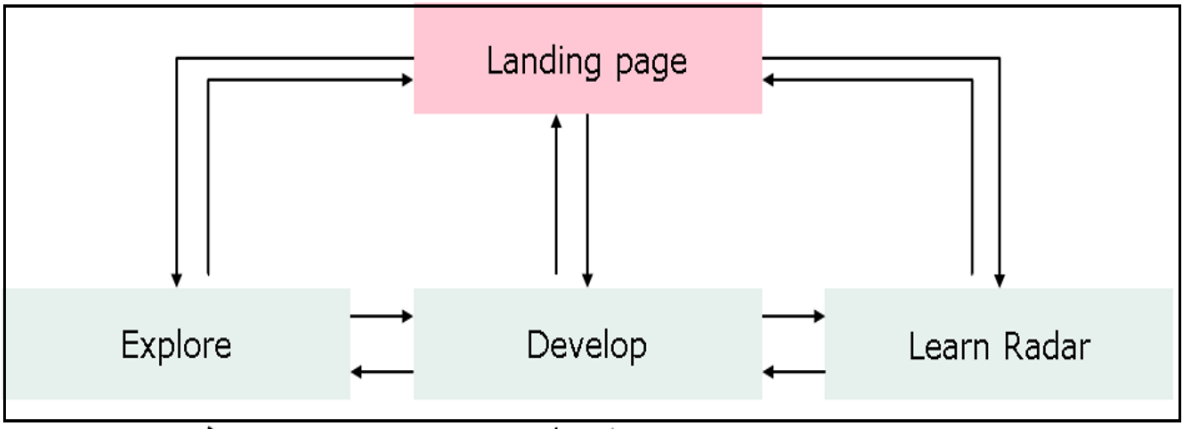
As mentioned before, the users of the webpage are the Infineon customers that's why having a good content without errors is fundamental.

2.1.2 Good quality of the images

In a webpage, good quality of images is a must.

2.1.3 Ability to navigates between pages

To get a good experience, we chose to adapt this architecture using a navigation bar for an easy accessibility.



2.2 *Functional needs*

The functional needs describe the key functionalities of the solution.

Our solution must provide the following functionalities:

- Ordering Infineon product.
- Ability to connect Sense2GoL board to the web page.
- Working applications with high performance.
- Arduino code for each application.

2.3 *Non-functional needs*

The non-functional needs cover all the requirements that we did not address in the functional needs. They specify the criteria that evaluate our website.

- Performance: the applications must provide the results in a timely manner.
- Reliability: the application must not crash.

3. Website conception

3.1 *Presenting the actors*

An actor is an external entity that explore our solution. This section describes the privileges of each actor. The user of our solution who can be Infineon customer, a partner or simple user. They will be able to order Infineon product (Sense2GoL demo kit) and test it is performance using the different kind of applications.

3.2 *Global use case diagram*

The diagram presented in Figure explains the global use case of our solution.

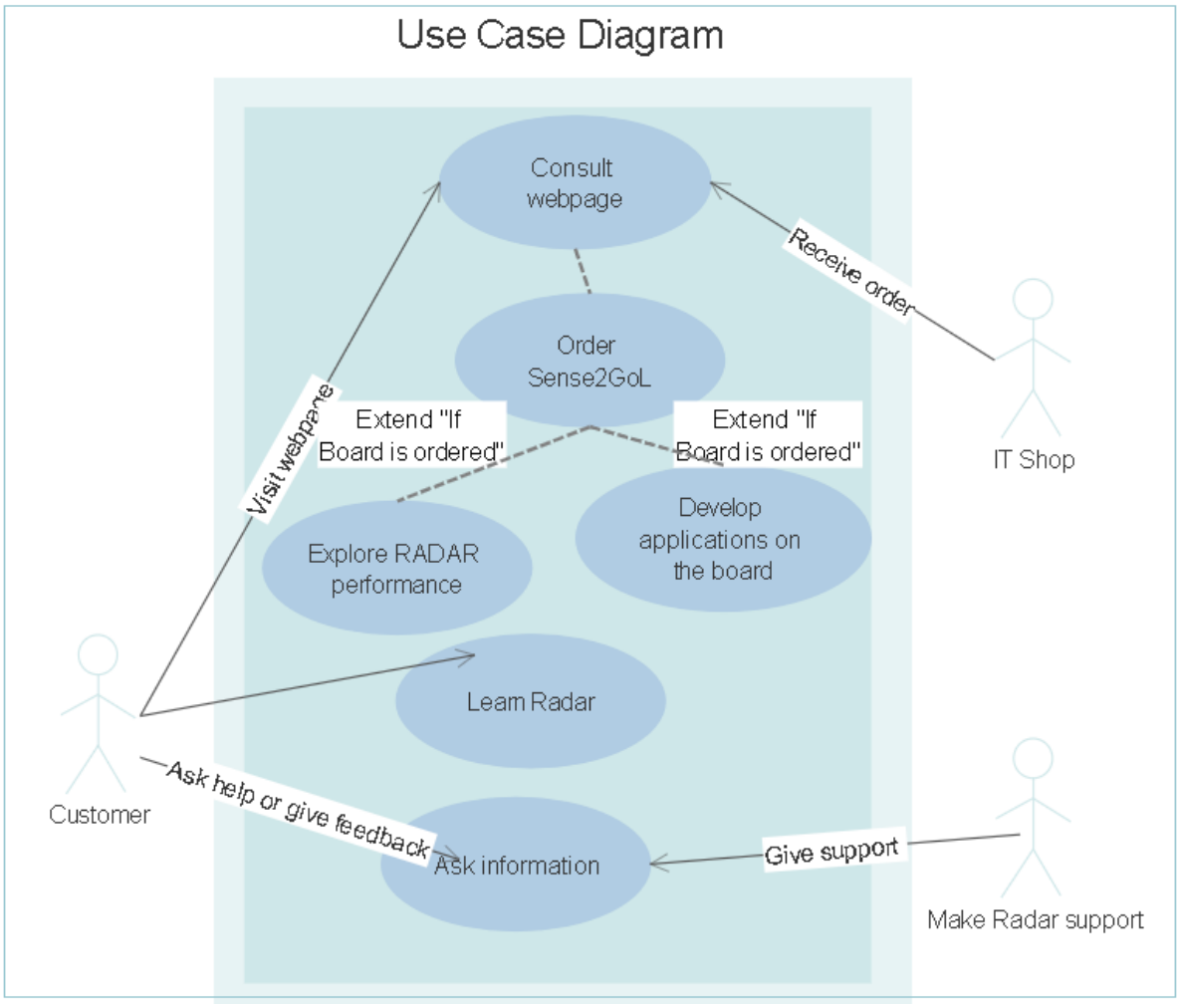


Figure 8: *Global use case diagram*

There are three ACTORS:

- The end user of the solution who is the customer, partner or simple user who will explore Infineon tutorial.
- The IT Shop, which are responsible of taking users order and the needed information for delivery.

Make Radar Support, who is taking customers feedback and helping them fix their problem if needed.

3.3 *System sequence diagram*

System sequence diagram use case “exploring part” once user got his product. The diagram below illustrates the web consulting stages made by the end user.

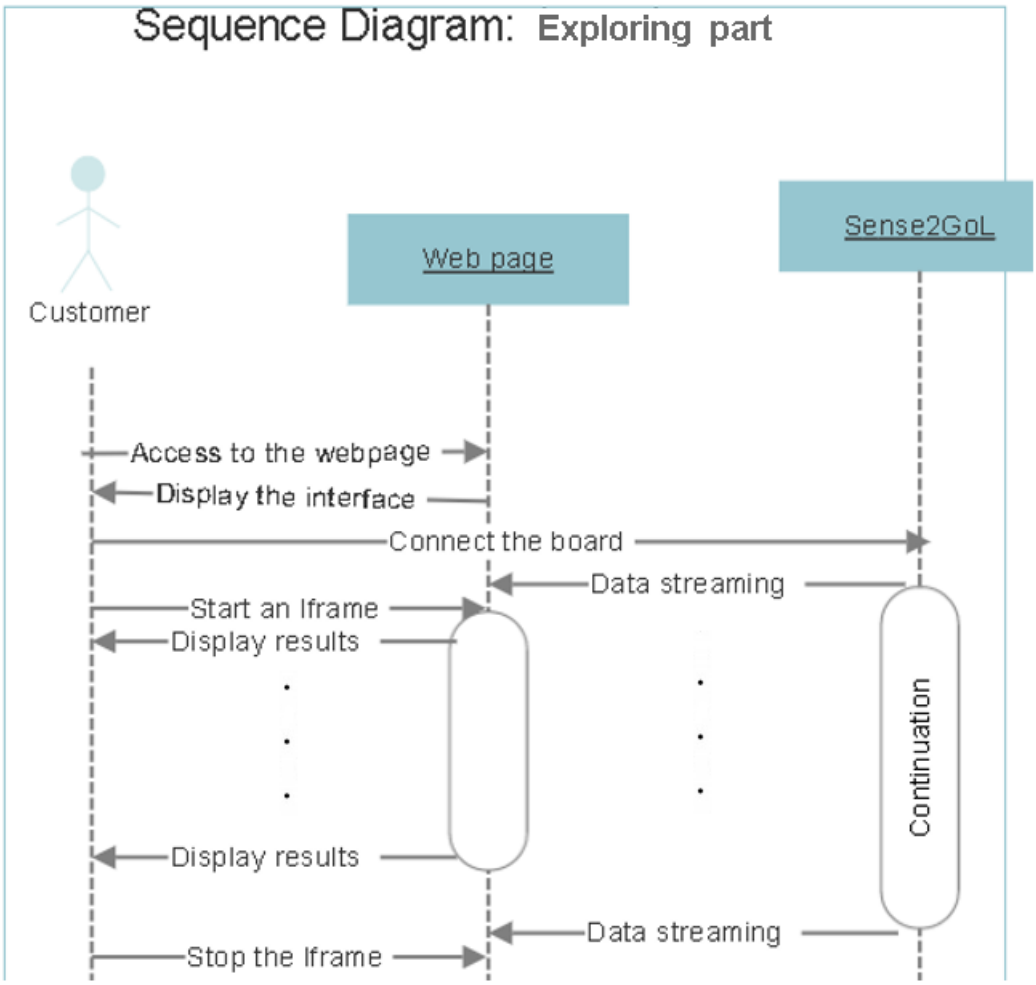


Figure 9: *Learning process*

The first step is getting through our webpage and then connect your board. Data start to flow once the radar is connecting. Starting an Iframe, will display the results until the user stops it.

3.4 *Refinement table of use case “Learning process*

Actor	End user
Precondition	Working iframes
Post-condition	Good learning experience
Nominal scenario	End user order the board and start the tutorial by exploring the radar performance, develop explored applications, understand the fundamentals behind radar

Table 4: *Refinement table of "Learning process"*

4. Website Services Specifications

4.1 *Landing page*

The landing page is the first thing that users see. It should be well designed with a summary of the educational tutorial, and an ordering button.

4.1.1 Design

As Make radar is a branch of Infineon website, the used template should the standard version and respecting the Infineon writing style.

4.1.2 Summarize

Summarizing the content will help customers to have a better understanding the learning approach and different levels

4.1.3 Ordering button

The ordering button is used to offer the possibility to buy the Infineon product. It connects users with the Infineon distrubuters.

4.2 *Explore part*

This chapter is created to test the Sense2GoL board on the user browser. By connecting the board using an USB, users have the possibility to access to different applications interactively.

4.3 *Develop part*

Open source codes are offered in this part to teach users how to develop radar application using a sample language which is Arduino. As users already explored the radar performance, they can now program those applications starting from the easiest application which is LED blink to more complicate application like presence detection.

4.4 *Learn Radar*

Learn radar is a chapter that explains the basics behind these application, the used techniques to calculate the speed and motion direction and signal processing process.

5. Radar Applications Specifications

5.1 *Radar Eqution*

This application is not to explore radar performance, but to teach one of the radar fundamentals, which is radar equation, followed by the time delay as the next equations show

$$Prx = \frac{Ptx\ Gtx\ Grx\ \sigma\ \lambda^2}{(4\pi)^2\ R^4}$$

Where :
Pt is the peak power transmitted by the Radar
G= Gtx=Grx G is the Antenna gain
σ is the Radar cross section of the target
R: Distance Target –Antenna
Λ: wave length

Equation 1: *Radar Equation*

$$Time = \frac{2D}{C}$$

Where :
D: Distance Target-Antenna
C: speed of light

Equation 2: *Time delay*

The radar equation application will help people understand how the radar works. Being able the change parameters (target, frequency, distance Target-Radar, transmitted power, gain), users will be able to understand the importance of each parameter in a cool and easy way.

6. Conclusion

This chapter has the purpose to show the conception of the webpage and the requirements. In the second half we gave an overview about the different applications to develop.

CHAPTER III: Design and Software Coding

1. Introduction

After putting the project in context, defining its key concepts and identify the functional, the nonfunctional needs and the analysis models then it will be followed with details about the development process of our applications.

2. Development process

Our proposed solution is a website containing different applications where customers can test the Infineon product simply by connecting their Sense2GoL Board via USB connection. Using Iframes, which are an HTML element that allows an external webpage to be embedded in an HTML document, many applications are presented on the same page.

IQ data are received by the radar and then transmitted to our applications where they are explored. Afterward, we apply our scripts that were developed in JavaScript. Finally, based on the purpose of each application, users can see and feel how the product works

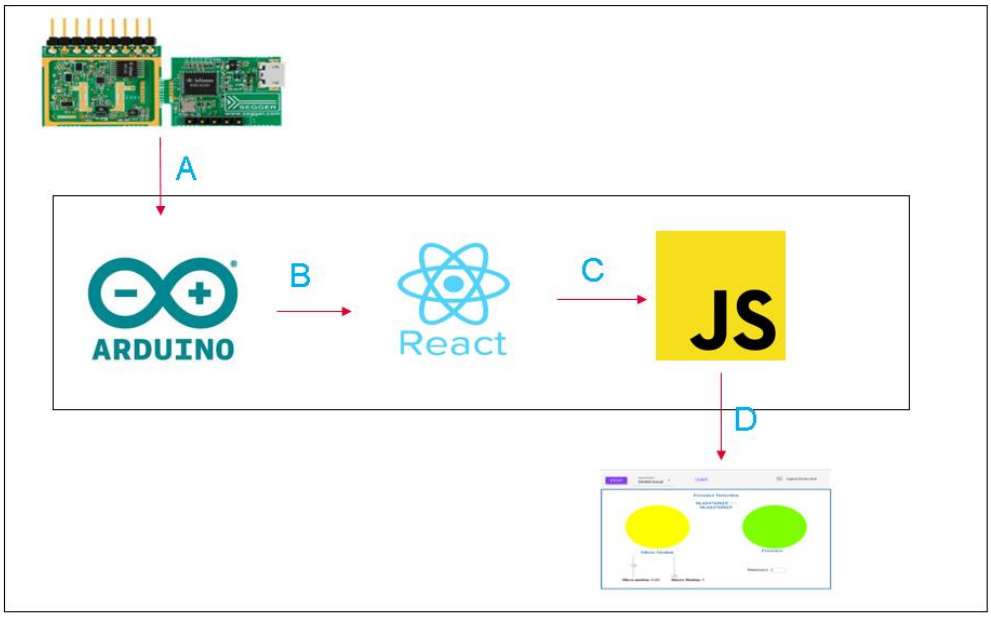


Figure 10: *Functional architecture*

The schema presents the different phases that our data will go through to achieve the desired results.

A: Once you connect the board, IQ data are gathered.

B: Arduino create agent is responsible for streaming data from our board to the browser.

C: Applying our JavaScript code.

D: Users interact with our applications by changing parameters and see how results change.

3. Technical architecture

3.1 Hardware environment

The hardware environment has the following technical characteristics:

Board	Frequency	Chip	Controller	Communication via
Sense2GoL	24GHz	BGT24LTR11	XMC1300 32-bit	USB

Table 5: *Board description*

Computer model	OS	Processor	RAM	Hard disk
LENOVO 20HAS1B20R	Windows 10 Entreprise	Intel® Core™ i5	8.00GB	256GB

Table 6: *Infineon computer configuration*

3.2 Software environment

We will define the development environments used for programming:

- OpenCMS: Web management
- React JS: JavaScript library
- Arduino Create Agent: listens to HTTP and Web Socket connections.
- Visual Studio code: A code editor developed by Microsoft

4. Iframes Architecture

In this part, we will share how our Iframes are built and the connection between different levels. Each one of them has this structure based on what we described in the second chapter

:

.

```
docs
  assets
    images
    iframes
      commons
        - bundle shared dependencies of the iframes
        - created by webpack
      section folders ...
        frame folders ...
        dist
          - bundled js by webpack
        src
          actions
            - services for communication with the daemon, fft, ...
          components
            - React components
          stores
            - mobx stores
            - data model
          app.js
            - React Entry Point
          app.scss
            - SASS imports
            - additional CSS
          index.html
            - entry point
            - loads the bundled dependencies from commons
            - loads the bundles js form dist
```

Figure 11: *Iframe architecture*

The next figure shows the different classes and how they communicate.

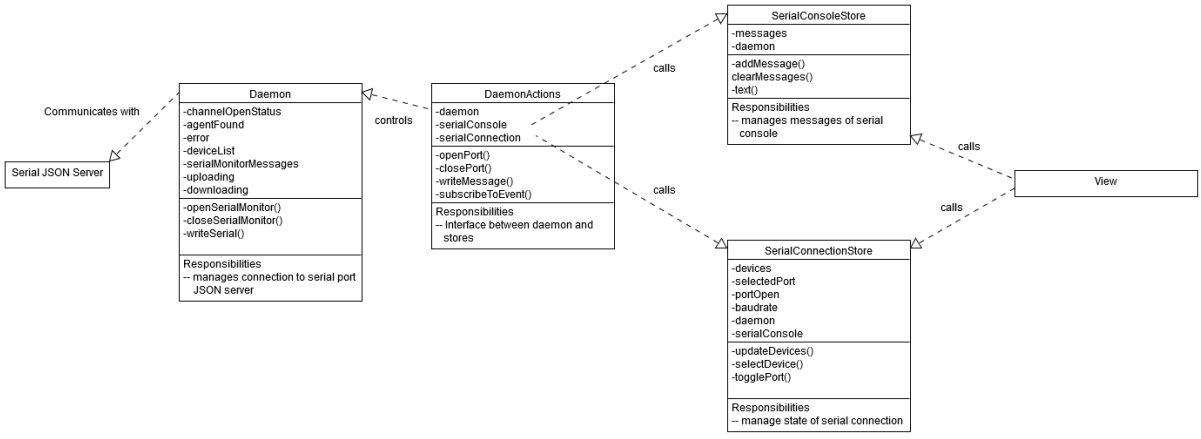


Figure 12: *Iframe Communication*

- Serial JSON Server is responsible of getting data from the radar.

- Daemon is an open-source library to communicate with Serial JSON Server.
- SerialConsoleStore and SerialConnectionStore are our developed JavaScript codes.
- DaemonActions applys our developed codes “Store” on the gathered IQ data by linking them.
- View is the composition of many developed components. It gathers outputs through SerialConsoleStore and SerialConnectionStore.

5. Presence detection

As the detection of human presence is becoming an important and even compelling requirement in a growing number of applications like, smart homes, healthcare, and security systems. We developed a demo application for our customers using Sense2GoL.

5.1 Web application

a. Overview

A presence detection application uses the IQ data generated from Sense2GoL board. Data is streamed from the board to our browser where we used a JavaScript algorithm. IQ data are classified as noise, micro or macro motion based on their magnitudes. We chose to use two kinds of motions (micro and macro motion) for many reasons:

The first one, macro motion is used to avoid false alarms caused by high levels of noise or in case something is moving near to the radar detection zone and we do not want to detect it. Macromotion is responsible for triggering the presence on.

Secondly, as our radar is highly sensitive, we classify breathing or typing as micro motion and we want to show that to our customers.

Current lighting solutions use timers; once there is no motion for a slot of time defined, we can assure that no one is present. We adopted that technique in our application. Whenever there is no motion for a few seconds (Time slot), we declare that no presence is here and we need a macro motion to trigger presence again.

b. Sequence diagram

In this part, we will explain the sequence diagram of our application.

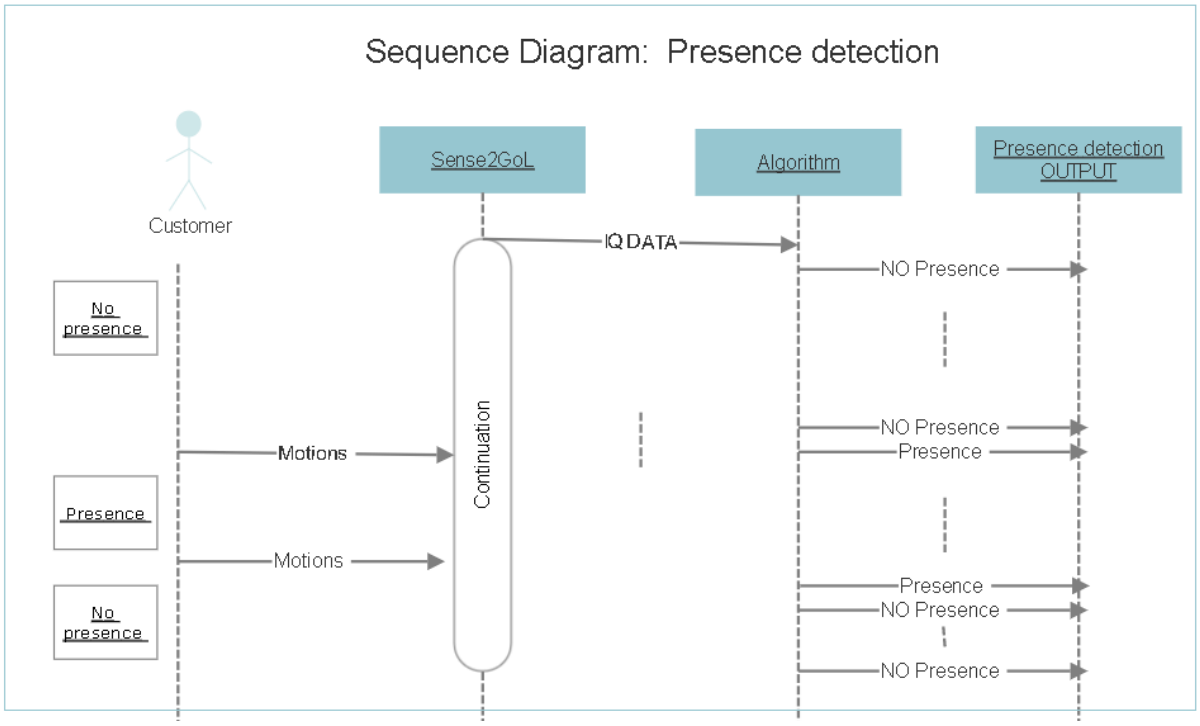


Figure 13: *Presence detection sequence diagram*

First, the radar sends IQ data. As there is no one in radar range the displayed message contains no presence. This stage stays the same until a macro motion is detected

Secondly, we keep detecting someone presence as long as we get motions. Finally, if the radar did not detect any micro motion during a time slot, then we declare the absence.

C. The developed application

This figure shows the developed application

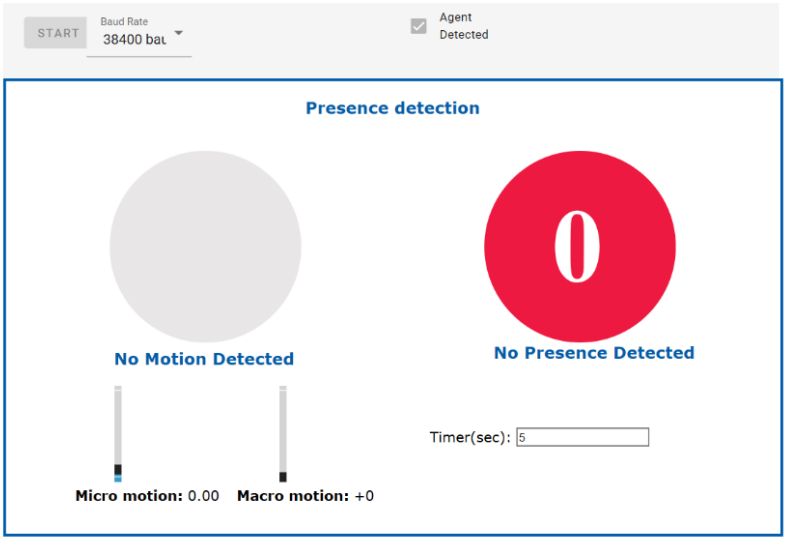


Figure 14: ***Presence detection interface***

This a dynamic application where users can interact with it by changing parameters:

- Micro motion: users can make their sensing more or less sensitive. By adjusting it, you can detect the tiniest motion like someone is breathing.
- Macro motion: as mentioned before, we use macro motion to avoid false alarms and the highest-level macro motion is set the bigger motion should be to trigger presence.
- Timer: Timer is the time slot. We reload the timer whenever there is micro or macro and in case no motion is detected after that time slot, we declare no presence is detected.

d. Demo

The next figure shows the different state of our application based on what was described in previous parts

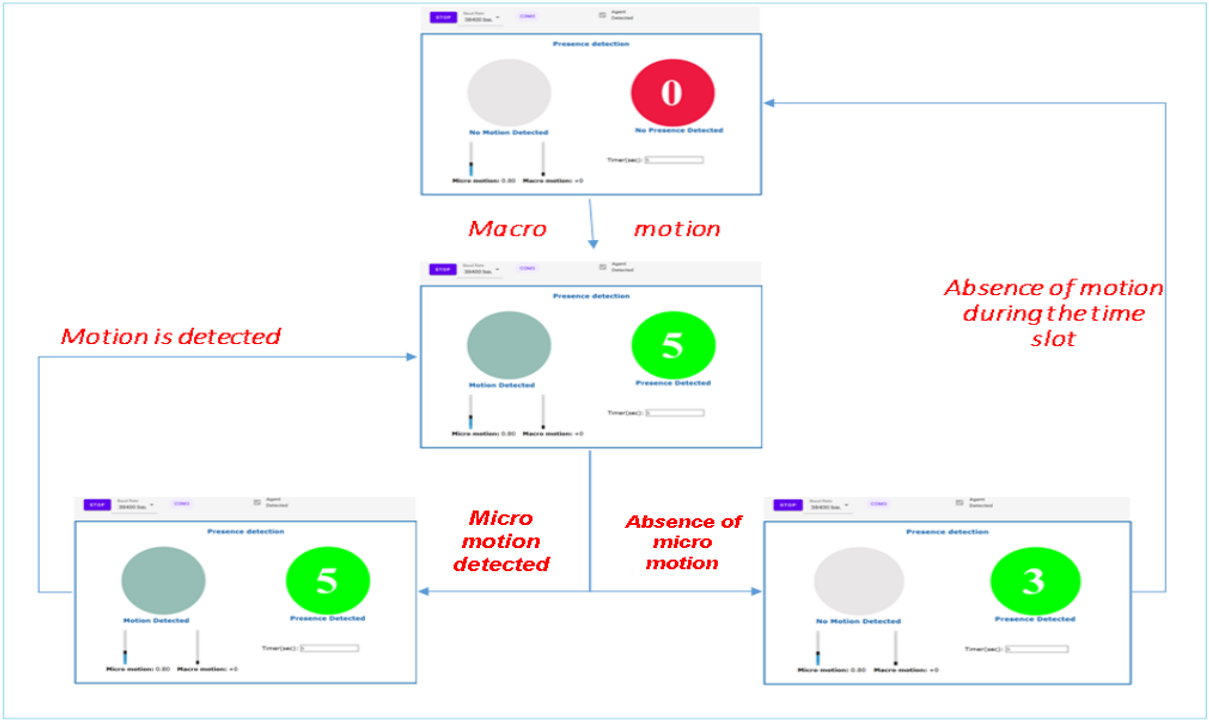


Figure 15: ***Presence detection demo sequence diagram***

e. Solution feedback.

- The advantage of our solution is that users can adjust the micro and macro motion threshold, which helps them to adapt it to their needs.

5.2 Presence detection using Arduino

5.2.1 Overview

As we mentioned in the first chapter, our tutorial is not about only exploring the radar performance, but also being able to develop this application with an easy language,

which is Arduino code. For that, we offered the source code written in an Arduino version with a description of each function

5.2.2 Demo

Once users develop their Arduino code, they will see results as the next figure shows.

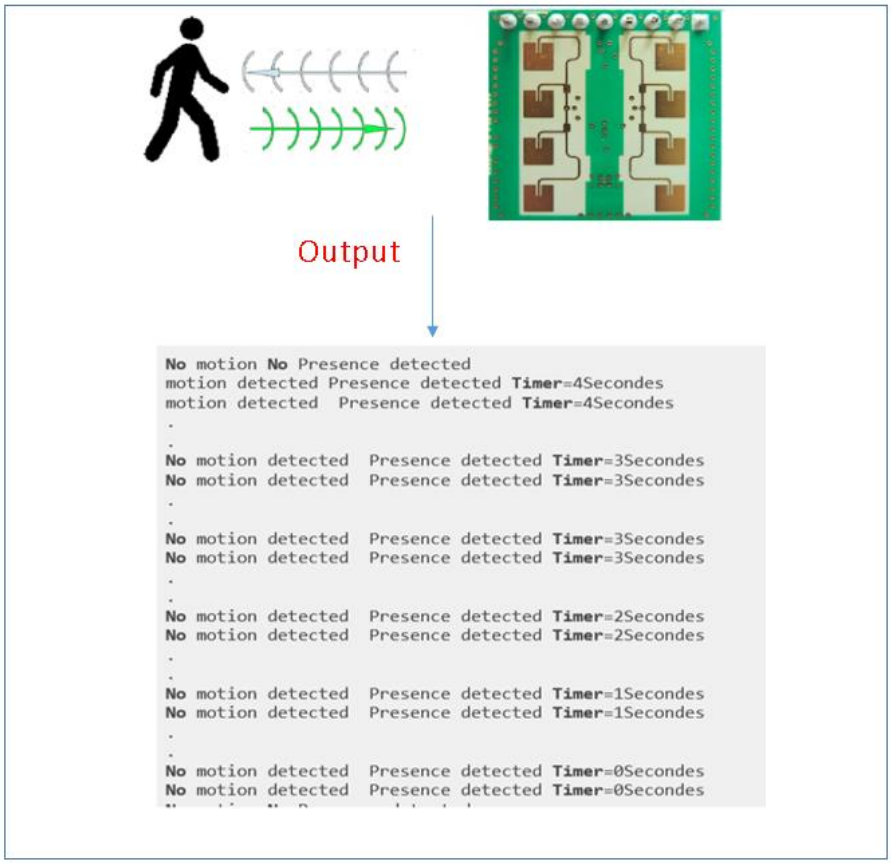


Figure 16: *Presence detection Arduino version*

5.2.3 Feedback

There is not difference between the web application feedback and the one with Arduino code.

5.3 Deep learning approach (extended)

5.3.1 Overview

Using deep learning for our solution was something we wanted to test and see if it can help us improve the performance using a high level of programming. For that, we created a dataset composed of two classed:

- Presence: 20164 samples of a person in front of the radar.
- Absence: 14710 samples of no presence in front of the radar.

5.3.2 Input data

For this approach, we did not use the standard input, which is IQ data. Instead of that, we used the FFT Spectrum of the IQ data. The purpose of using the FFT spectrum is to extract new features different from what we already used.

The next figure describes how to get FFT Spectrum (Input data) from IQ data

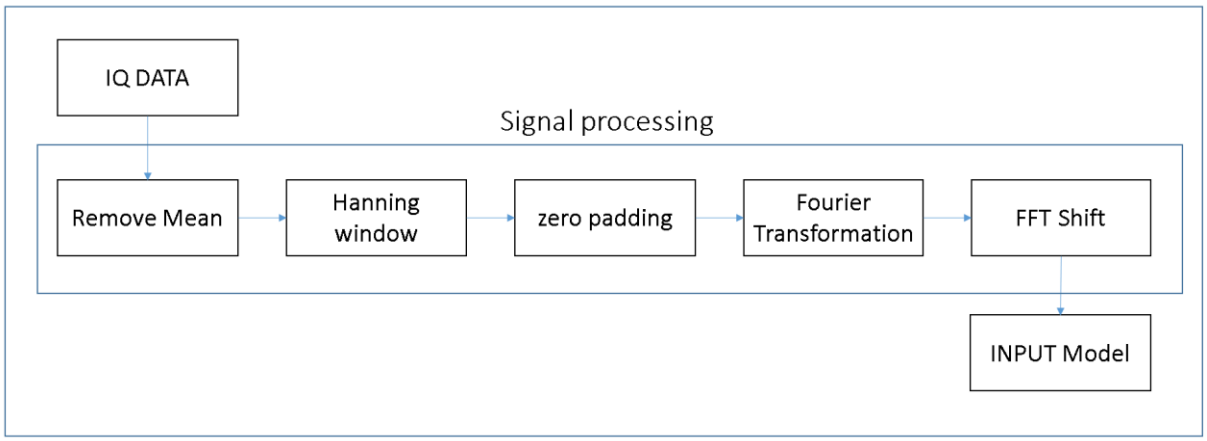


Figure 17: *Signal processing*

Now, we will describe each technique used in our algorithm:

- IQ Data: Our input is a buffer of 128 values for each I Data and Q data.
- Remove mean: This method is used to normalize the range of independent variables or features of data.
- Hann window: To reduce the amplitude of the discontinuities at the boundaries of each finite sequence acquired
- Zero padding: Zero padding is used to compute heavily interpolated spectra by taking the DFT of the zero-padded signal. Such spectral interpolation is ideal when the original signal is time limited (nonzero only over some finite duration spanned
- by the orignal samples). Our padding coefficient equals to 3 and that gave us a buffer of 512 values for each.
- FFT Transformation: To calculate the discrete Fourier transform (DFT) of a sequence
- FFT Shift: To rearrange a Fourier transform signal by shifting the zero-frequency component to the center of the array

The next figure shows our input data and output data after applying signal-processing techniques:

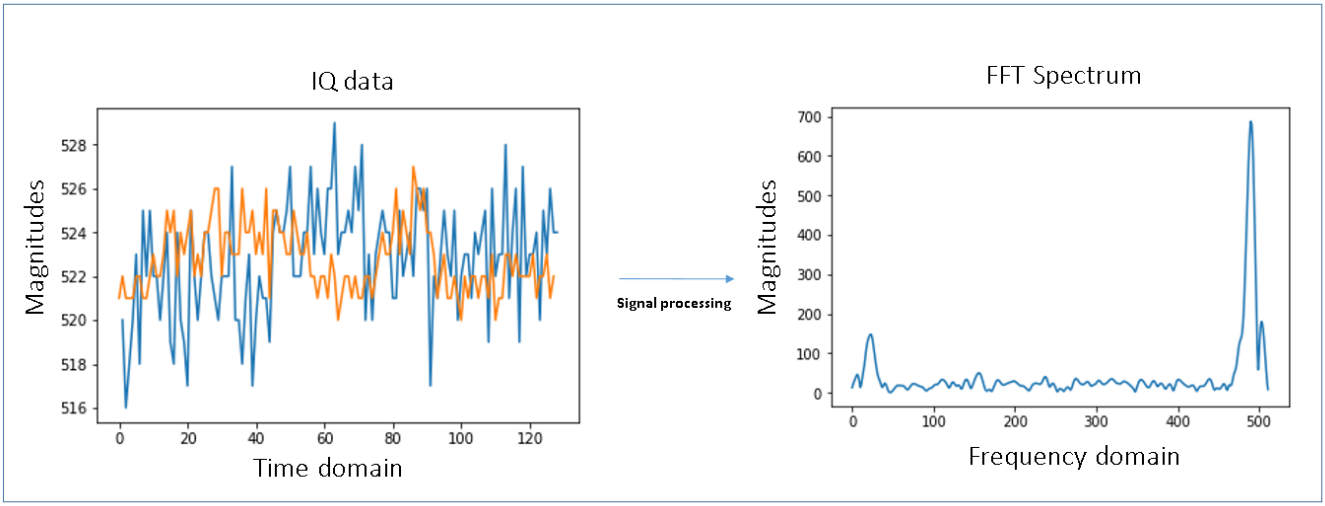


Figure 18: *From IQ data to FFT Spectrum*

5.3.3 1D CNN

Convolutional Neural Network (CNN) models were developed for image classification, in which the model accepts a two-dimensional input representing an image's pixels and color channels, in a process called feature learning [17].

The same process applies to one-dimensional sequences of data. The model extracts features from sequences data and maps the internal features of the sequence. A 1D CNN is very effective for deriving features from a fixed-length segment of the overall dataset, where it is not so important where the feature is located in the segment.

The reasons for using this model are:

- It analysis frequency domain data.
- It uses large filter sizes
- It analysis data with a fixed-length period

5.3.4 Model Architecture

VGG16 Architecture inspires our proposed model. This architecture is known as "Very deep convolutional Networks for Large-Scale Image Recognition" but in our case, we are using it with 1D input.

The next figure shows are structured model:

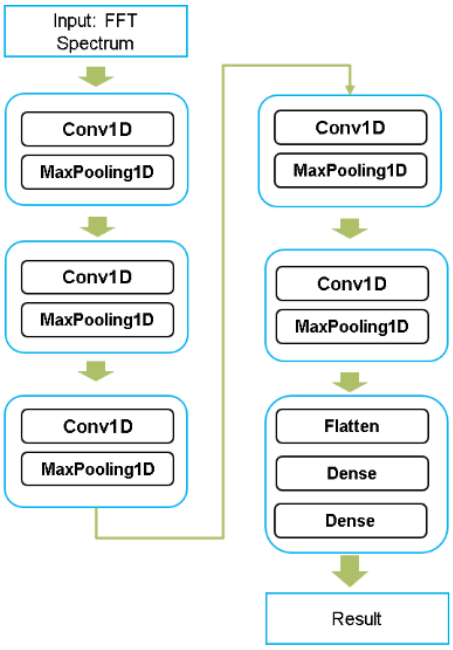


Figure 19: *1D CNN Architecture*

5.3.5 Results

For our solution, we used five hidden layers and different activation function.

The next table shows the different results that we got after applying different activation function.

Activation function	Relu	Tanh	Softmax	Sigmoid
Accuracy	0.427	0.572	0.941	0.939
Loss	0.693	0.693	0.171	0.177

Table 7: ***Activation function comparison***

- Accuracy: Accuracy is one metric for evaluating classification models.

- Loss model: The loss is calculated on **training** and **validation** to know how well the model is doing for these two sets.

⇨ through this table, we can conclude that the best activation functions with higher accuracy and less loss are Softmax and sigmoid

5.3.6 Softmax

After testing different activation functions, I chose to use softmax as activation function and the next images shows the mentioned performances:

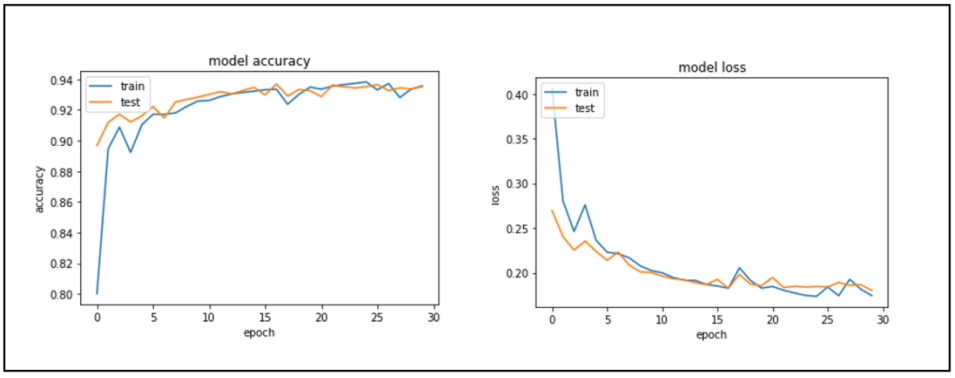


Figure 20: *Model accuracy - Loss function*

Our figure compares the performance of our model based on the training dataset and test dataset.

confusion matrix	Predicted (Absence)	Predicted (Presence)
Actual (Absence)	2951	442
Actual (Presence)	31	3551

Table 8: *Confusion Matrix*

Confusion matrix evaluation:

- true positives (TP): 2951
- true negatives (TN): 442
- false positives (FP): 31
- false negatives (FN): 3551

⇨ Based on the matrix evaluation, our model is subtitle to give us false alarms due to high values of (TN):

⇨ To avoid false results caused by (TN), we added a counter that takes decision based on a sequence of outputs:

- After getting five consecutive FN, we display a presence is detected.
- After getting ten consecutive TP, we display that no presence is detected
- We keep the same state if none of these conditions is true.

⇨ After adding the counter, the performance of our model improved

5.3.7 Demo

This part shows the output of our model based on one's presence:

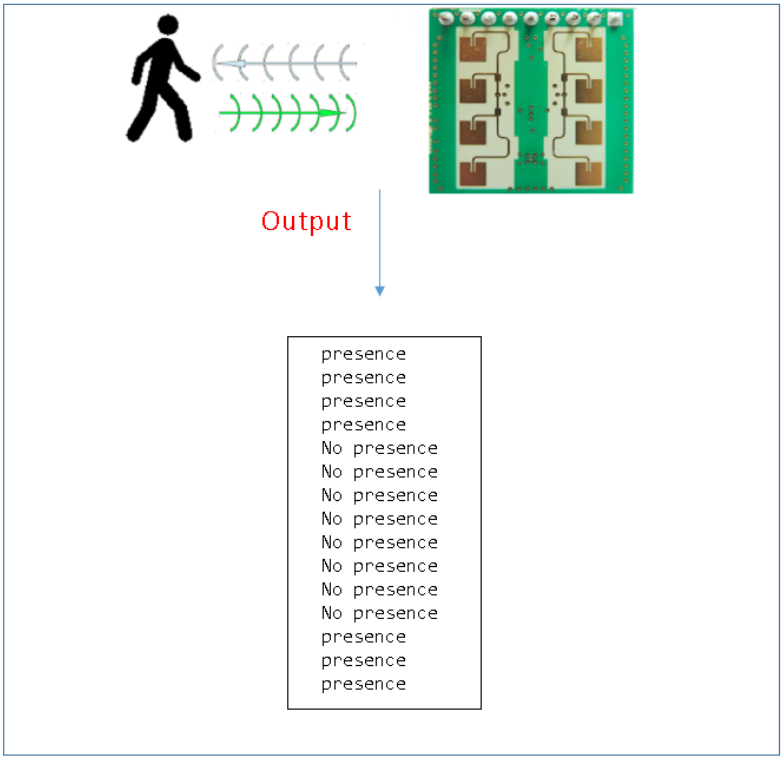


Figure 21: *Presence detection (Deep learning model)*

5.3.8 Feedback

The positive side of using this approach is avoiding adjusting the threshold to get good results. However, the negative sides are:

- Even with good accuracy, someone in front of the radar without breathing is still hard to detect it.
- There is no concrete feature to be interpreted that user can explore.

6. Radar equation

6.1 Developed application

In this part, we will describe the developed application and how users can interact with is. The next figure shows the developed interface

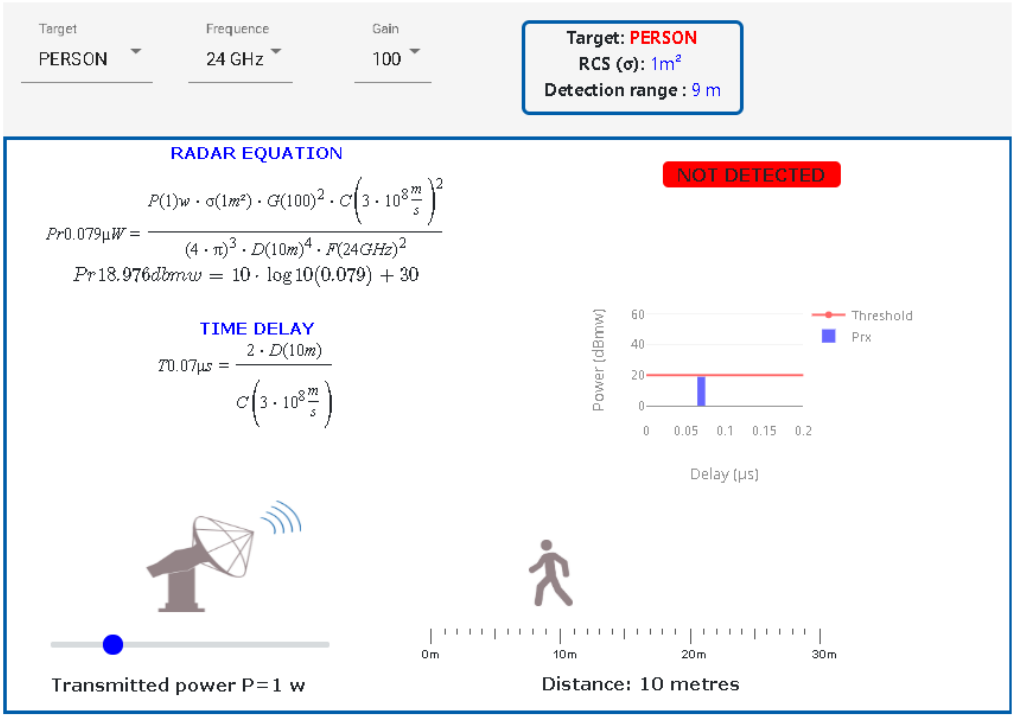
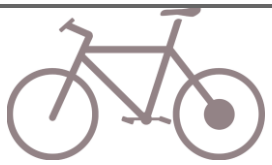


Figure 22: *Radar Equation Interface*

- **Changeable parameters:**
  - Users can choose one of four targets (person, bicycle, car and an airplane). Changing from one target to another will change the RCS value as the table shows:

Table 9: *Targets & RCS*

Person	Bicycle	Airplane	Car
			
RCS = 5 m^2	RCS = 5 m^2	RCS = 30 m^2	RCS = 150 m^2

  - Frequencies: 24GHz or 60GHz.
  - Gain: 10, 100 or 300.
  - Transmitted power: between 0 Watt to 5 Watt.
  - Distance Target –Antenna: Dragging the object to the left or to the right will increase or decrease the distance between the radar and target

- **Changeable output**

- The received power: The value of received power is dynamically calculated whenever a parameter is changed.
- The detecting range is updated whenever there is a change of transmitted power, gain, frequency, and target.
- Time delay: reflects the time that a transmitted wave takes to hit the target and come back as received wave, which depends on the distance target-antenna.
- Detected- Not detected: Reflects, if our target can be detected or not based on the minimum detectable signal.
- Power/Delay plot: this plot reflects the received power and the delay.

7. Conclusion

In this chapter, we have defined our functional architecture. We have shown our developed applications with different approaches and used techniques.

Chapter IV: Solution implementation

1. Introduction

This final chapter presents the final phases of the project that consists of organizing all the contents.Some applications are shown in this chapter but not described before. A working student before me built those applications.

2. Advertisement

To advertise our webpage and product, we shot a video which will be uploaded on YouTube. A dialog between Ismail NASR, who is product marketing for industrial Radar and Mohamed JMOUR, the developer of Make radar. The purpose of the video is to give an overview of our webpage and how users can take benefits from it.

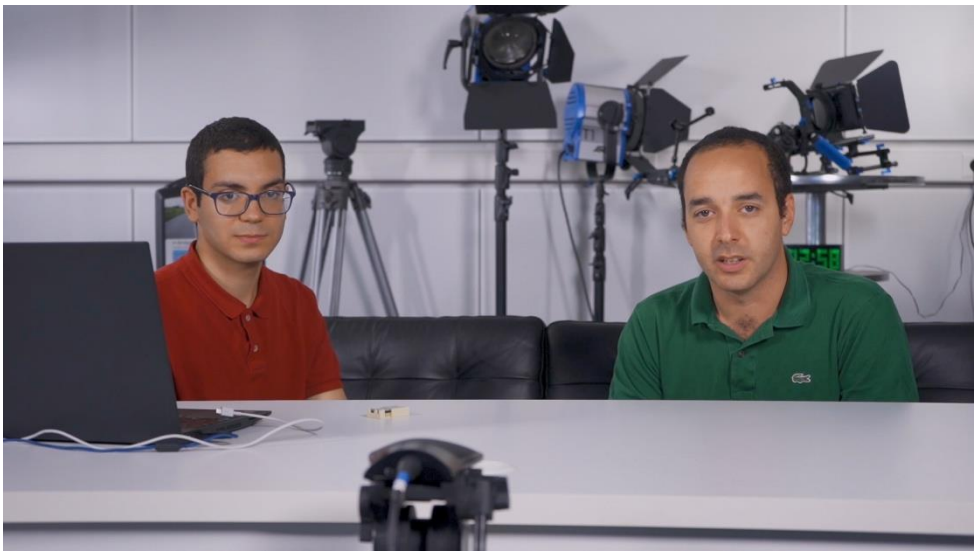


Figure 23: *Make Radar (video)*

3. Landing page

Our landing page is the first page that users will see it. Briefly describes each part of the tutorial and importantly ordering Infineon product

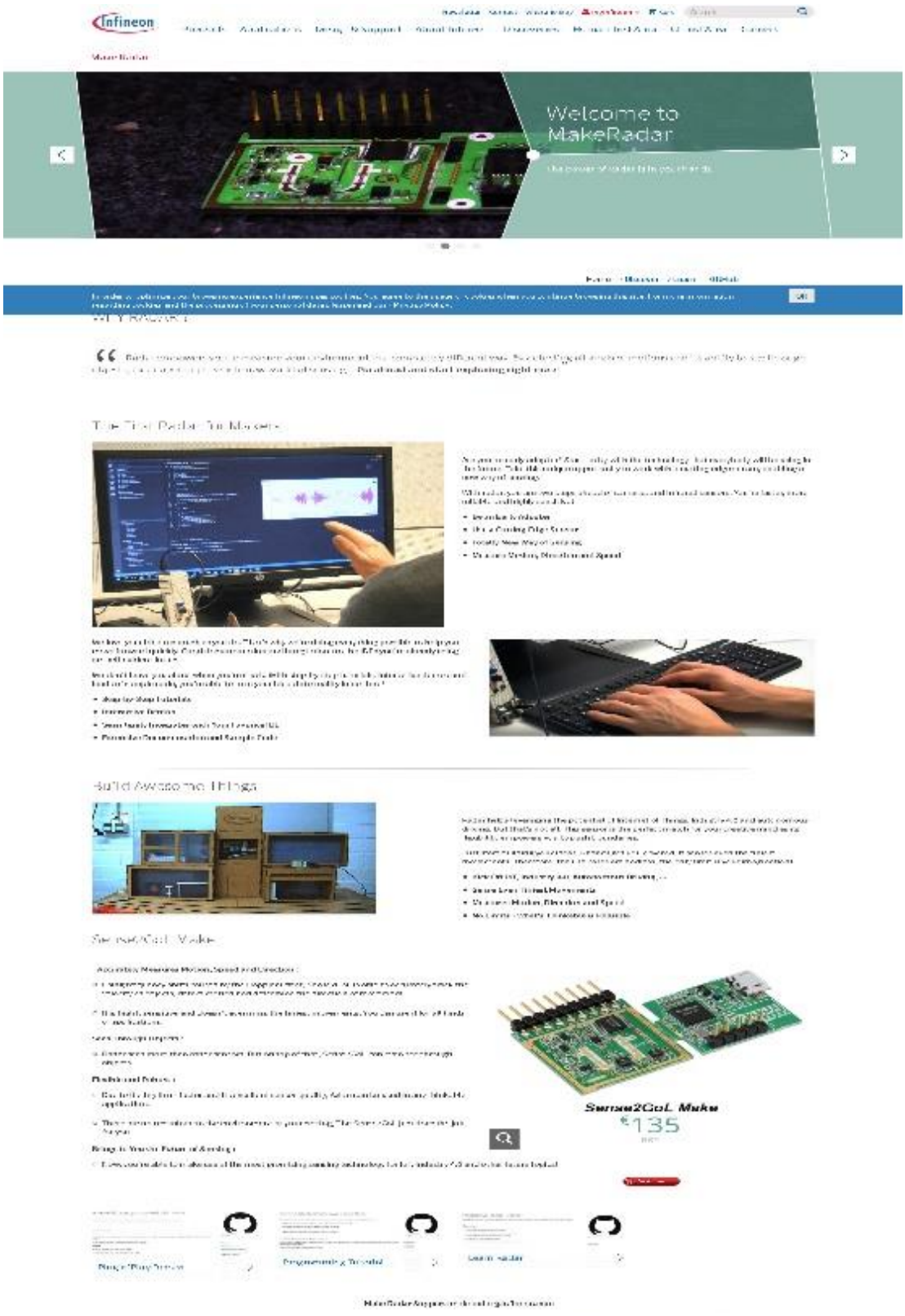


Figure 24: Landing page

4. Plug'n'Play (Explore)s

4.1 before you start

Presents the needed steps to start exploring the radar performances.



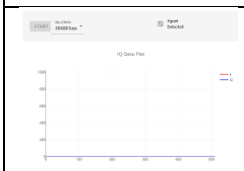
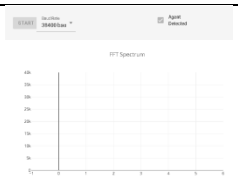
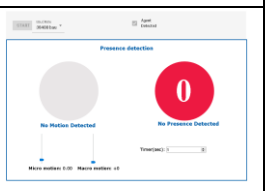
Figure 25: Setup

4.2 Explore applications

This part offers different application to be tested

The next table shows all the applications in that user can test:

Table 10: ***Applications to be explored***

Raw data	speedometer	FFT Spectrum	Door Opener	Presence detection
				
This Iframe shows the received IQ data and where users see their motion detected.	**This application allows you to test the speed of motions.**	**FFT Spectrum plots the Spectrum of IQ data**	**Door opener detects your direction of movement. If you are approaching then it will open for you.**	**Detect some one's presence in the radar range**

5. Develop part

The develop part provides all the used codes to develop those applications in Arduino version. Learning to develop Arduino from level zero like LED blink to more complicated application like speedometer and presence detection.

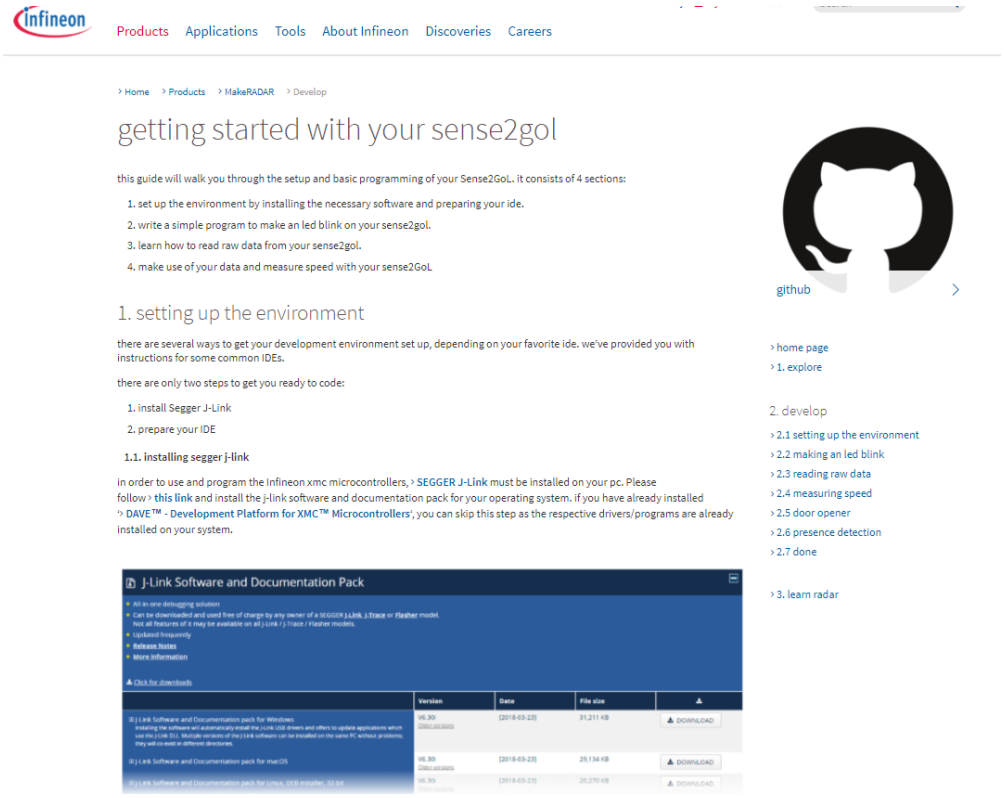


Figure 26: Develop part

2. making an led blink

to make an led blink on your sense2GoL, there are just two steps necessary:

1. set-up the board
2. let the led blink

2.1. set-up the board

for the `setup()` method, we only have to set the `LED1` pin to `OUTPUT`, so we can control the led:

```arduino
void setup() // This code is executed once for the initial setup
{
  pinMode(LED1, OUTPUT); // Set our LED as an output, so it can be controlled

}
```

2.2 let the led blink

Figure 27:LED BLINK

6. Learn Radar

This part contains the fundamentals of radar starting by Electromagnetic Waves, radar equation and more.

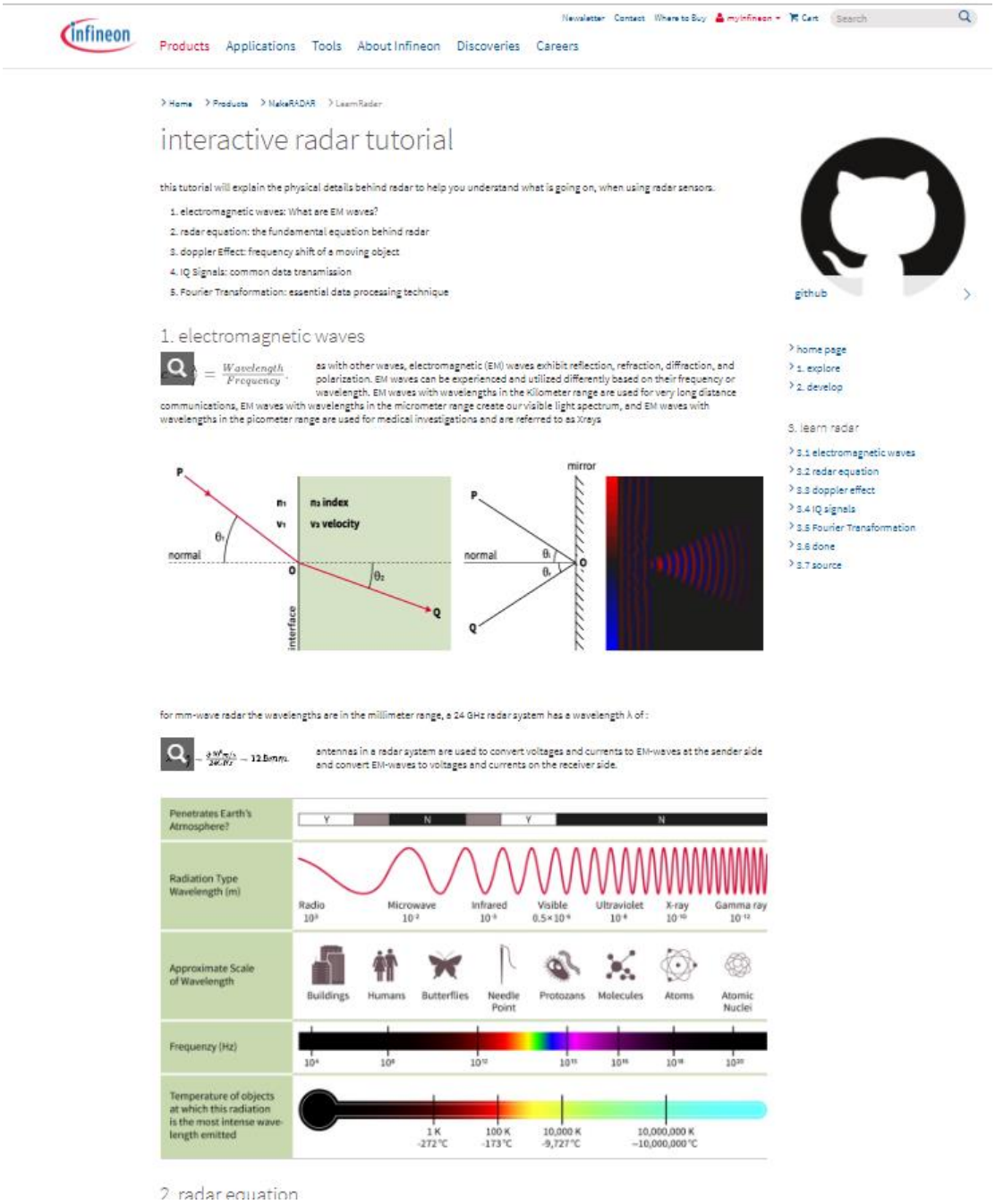


Figure 28: Learn Radar

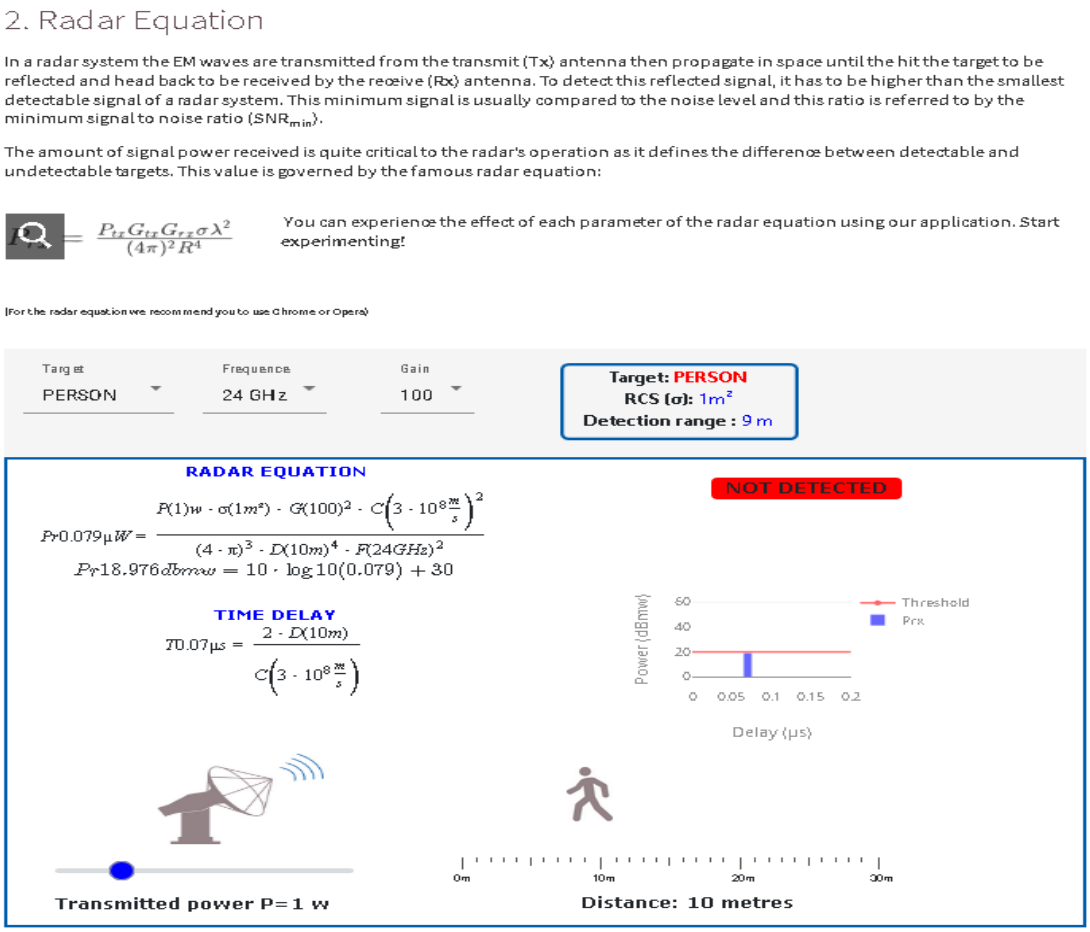


Figure 29: *Learn Radar equation*

Then you can find an explanation of the Doppler Effect and how velocity is calculated, and we finish with Fourier Transformation. We are using an open-source application that interactively explains Fourier transformation

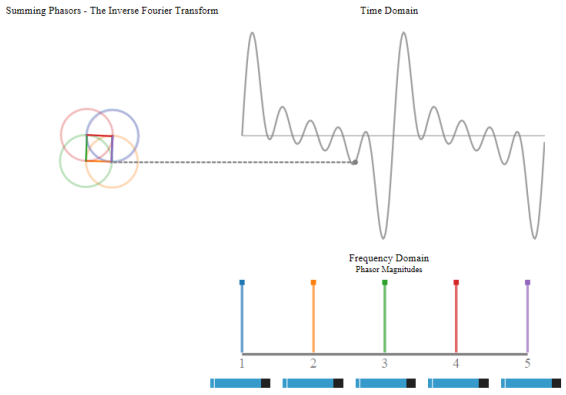


Figure 30: *Fourier Transformation [18]*

7. Development Notes

This page is dedicated to developers who worked on building the website, where they share their notes and used tools for their successor.

8. Conclusion

In this chapter, we showed an overview of our website, which is destined to Infineon customers, partners, sample users to see and test the Infineon product.

General conclusion and future works

The first objective of our project is to build a website, which not only explains radar interactively and playfully, but also gives users a hands-on experience as they can run several applications directly from their browser. For our development, we relied on React JS structure to build different and interactive Iframes that shows what our board is capable of and how accurate it is. The second part focuses on explaining step-by-step how to build your first radar project on Arduino. The final part explains the basics of the radar in a sample and easy way.

There are several things to do in the future to improve our website: Not only add new applications and Arduino code but also add new Infineon boards, more advanced like Distance2Go and Position2Go for distance and angle detection. After adding new products, customers will have a full package of Infineon products and they can choose the perfect one for their uses case.

This project has been very beneficial in theory and practice. It allowed us to consolidate our knowledge as a developer and marketer person. Then it enriched us our sense of synthesis through creating content and test its use, it allowed us to deepen our practical knowledge to be able to solve the problems encountered.

Finally, we should also mention that this project has allowed us to unleash our creativity and test our determination to solve problems, and our exchange capacity and interaction with the company’s colleagues that have prepared us for the professional life.

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